



**Stan's
Technologies**

Benchmark Report

BoolMin2D vs PyEDA Boolean Minimization

Experiment Date: 2026-01-13

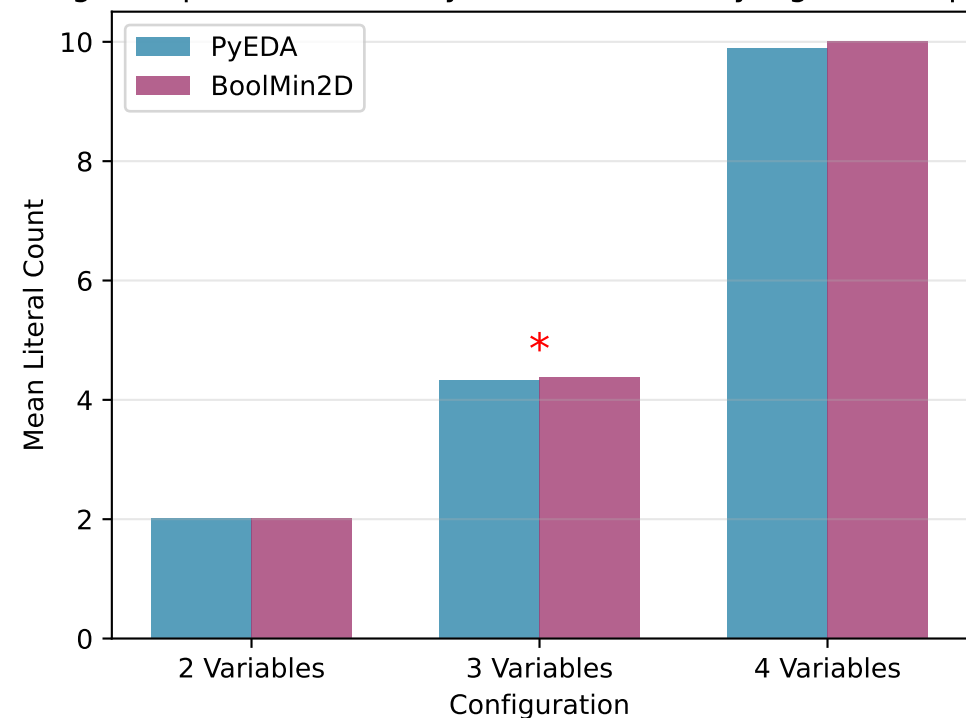
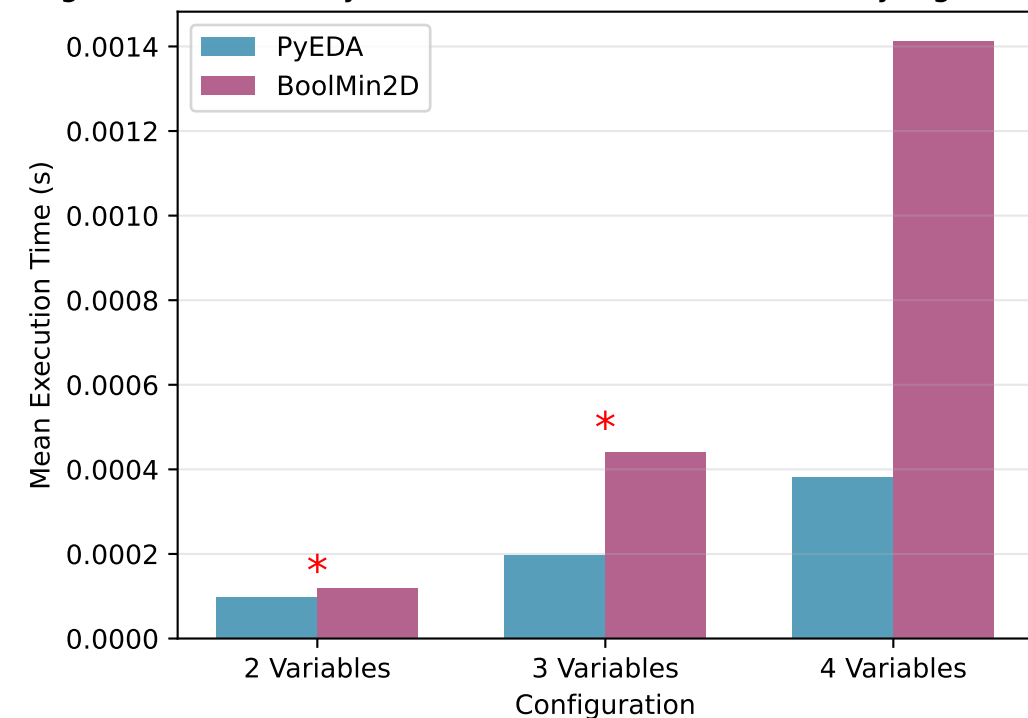
Random Seed: 42

Total Test Cases: 3000

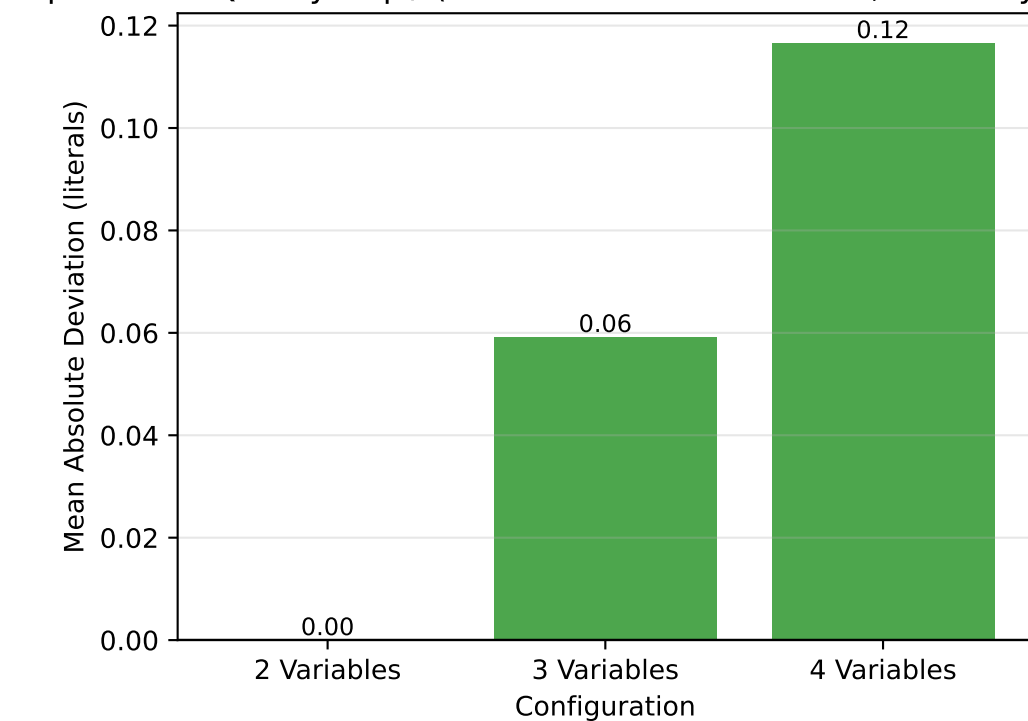
Statistical Significance Level: $\alpha = 0.05$

A Statistical Comparison of Boolean Minimization Algorithms

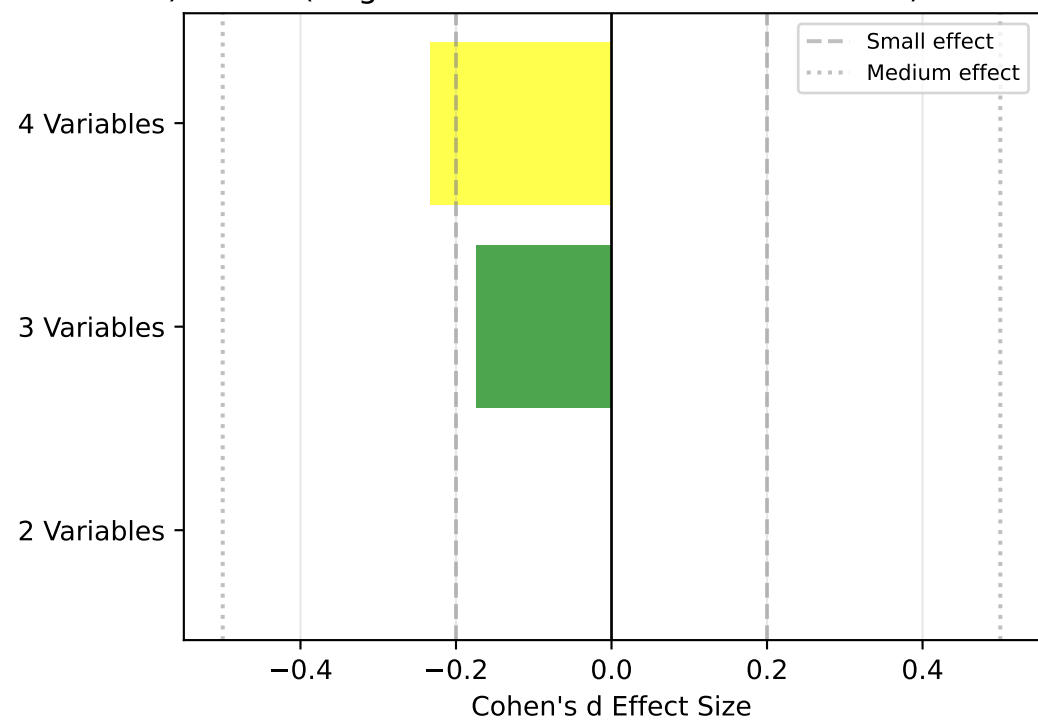
Average Performance by Variable Count\n(* = statistically significant, p < 0.05)



Simplification Quality Gap\n(Green = BoolMin2D better, Red = PyEDA better)



Effect Size: Simplification Quality
(Negative = BoolMin2D more minimal)



STATISTICAL ANALYSIS SUMMARY

EXPERIMENT OVERVIEW

=====
Total Test Cases: 3000
Configurations Tested: 3 (2, 3, 4 variables)
Tests per Config: 10 (sparse, balanced, dense distributions)
Random Seed: 42
Significance Level: $\alpha = 0.05$

2-VARIABLE FUNCTIONS (1000 tests, 747 non-constant)

EXECUTION TIME:
Mean PyEDA: 0.000099 s
Mean BoolMin2D: 0.000120 s
Mean Difference: -0.000021 s
95% CI: [-0.000027, -0.000015]
p-value: 0.000000
Statistically Sig: YES
Effect Size (d): -0.2302 (small)
Winner: PyEDA

SIMPLIFICATION QUALITY (Literal Count):
Mean PyEDA: 2.02 literals
Mean BoolMin2D: 2.02 literals
Mean Difference: +0.00 literals
Mean Deviation: 0.00 literals
Quality Gap: 0.00% relative to PyEDA
95% CI: [nan, nan]
p-value: nan
Statistically Sig: NO
Effect Size (d): 0.0000 (negligible)
Winner: TIE

3-VARIABLE FUNCTIONS (1000 tests, 946 non-constant)

EXECUTION TIME:
Mean PyEDA: 0.000196 s
Mean BoolMin2D: 0.000442 s
Mean Difference: -0.000246 s
95% CI: [-0.000258, -0.000233]
p-value: 0.000000
Statistically Sig: YES
Effect Size (d): -1.2240 (large)
Winner: PyEDA

SIMPLIFICATION QUALITY (Literal Count):
Mean PyEDA: 4.33 literals
Mean BoolMin2D: 4.39 literals
Mean Difference: -0.06 literals
Mean Deviation: 0.06 literals
Quality Gap: 1.37% relative to PyEDA
95% CI: [-0.08, -0.04]
p-value: 0.000000
Statistically Sig: YES
Effect Size (d): -0.1746 (negligible)
Winner: PyEDA

4-VARIABLE FUNCTIONS (1000 tests, 995 non-constant)

EXECUTION TIME:
Mean PyEDA: 0.000382 s
Mean BoolMin2D: 0.001412 s
Mean Difference: -0.001030 s
95% CI: [-0.001061, -0.000999]
p-value: 0.000000
Statistically Sig: YES
Effect Size (d): -2.0742 (large)
Winner: PyEDA

SIMPLIFICATION QUALITY (Literal Count):
Mean PyEDA: 9.89 literals
Mean BoolMin2D: 10.01 literals
Mean Difference: -0.12 literals
Mean Deviation: 0.12 literals
Quality Gap: 1.18% relative to PyEDA
95% CI: [-0.15, -0.09]
p-value: 0.000000
Statistically Sig: YES
Effect Size (d): -0.2332 (small)
Winner: PyEDA

CONCLUSIONS

KEY FINDINGS

- 1. PERFORMANCE:
 - Average PyEDA time: 0.000226 s
 - Average BoolMin2D time: 0.000658 s
 - Difference: -0.000432 s
 - Statistical significance: YES (p = 0.000000)
- 2. SIMPLIFICATION QUALITY:
 - Average PyEDA literals: 5.75
 - Average BoolMin2D literals: 5.81
 - Mean deviation: 0.06 literals
 - Quality gap: 1.11% relative to PyEDA
 - Statistical significance: YES (p = 0.000000)
- 3. EQUIVALENCE:
 - All 3000/3000 tests passed logical equivalence checks
 - Both algorithms produce functionally correct results
- 4. EFFECT SIZES:
 - Performance: -0.8092 (large)
 - Simplification: -0.1741 (negligible)

INTERPRETATION

Per-Variable Analysis:

- 2 variables: 0.00% quality gap, Neither produces more minimal results (Mean deviation: 0.00 literals)
- 3 variables: 1.37% quality gap, PyEDA produces more minimal results (Mean deviation: 0.06 literals)
- 4 variables: 1.18% quality gap, PyEDA produces more minimal results (Mean deviation: 0.12 literals)

RECOMMENDATIONS

Based on this statistical analysis:

- Both algorithms achieve statistically equivalent simplification quality
- Algorithm choice should be based on other factors (API, integration, etc.)

REPRODUCIBILITY

- Random seed: 42
- All test cases can be reproduced with this seed
- Statistical tests use standard methods (paired t-test, Cohen's d)